

HPS Gaussian-Process Resonance Search

Reconstructed-MC Injection and Signal Recovery Standalone Study v6.2

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Abstract

Signal extraction is tested by adding exactly 1,000, 5,000, 10,000 and 30,000 selected candidates from native smeared 2021 MC distributions to 40 background toys at each mass from 60 through 260 MeV in 20 MeV steps. Paired fits compare windows centered at the generated mass and the reconstructed MC core, with the same nominal width. At 30,000 injected candidates, subtracting the paired zero-injection fit gives a mean recovered increment of 0.715–0.895 of the injected yield for the original window and 0.780–0.938 for the shifted window over 60–240 MeV. The shift reduces the loss but does not restore full recovery. Controls with signal removed from the training sidebands attribute much of the deficit to the GP response to the injected tails. Pulls and profile likelihood inclusion counts accompany the yield tests.

1 Result and relation to v6.1

The selected-candidate MC distributions examined in v6.1 have broad tails. Injected candidates outside the fitted window enter the sideband sample and can change the GP prediction beneath the signal core. The recovery tests include this response to the injected counts.

Injected N	Pole-centered	MC-core-centered
1,000	-0.089–5.227	-0.069–5.566
5,000	0.689–1.737	0.740–1.820
10,000	0.783–1.301	0.837–1.372
30,000	0.822–1.011	0.845–1.073

Table 1: Ranges of the mean fitted yield divided by the injected yield over 60–240 MeV. These summarize the ten mass points; they are not confidence intervals. The low-injection ratios include the background offset and should be read with their toy errors in Figure 1.

The tests use fixed empirical MC shapes and one background mean. The v6.1 audit did not establish signal-daughter association or full equivalence between the MC and data selections. The template and its MC-only center are held fixed; finite-MC, background-truth and detector-response uncertainties are not sampled.

2 Toy ensemble and yield normalization

The GP arithmetic mean fitted over the full mass support of the 2021 10% spectrum in v5.8.2 defines the background expectation μ_i over the unchanged analysis support $D = [36, 299.75]$ MeV. Independent Poisson spectra are generated for each mass and toy index. At a fixed mass, the same spectrum is used at all four injection levels and for the zero-injection fit. The two window prescriptions are fitted to the same injected spectrum.

The signal probabilities p_i are obtained from the native selected-MC histogram at that generated mass. They use the full selected normalization, including the recorded overflow. Counts are drawn as

$$B_i \sim \text{Poisson}(\mu_i), \quad (S_1, \dots, S_K, S_{\text{below}}, S_{\text{above}}) \sim \text{Multinomial}(N; \mathbf{p}), \quad n_i = B_i + S_i. \quad (1)$$

The last two categories retain candidates outside D , so the total draw is exactly N even when fewer candidates enter the analysis. CDF rebinning treats the density as uniform within each source histogram bin. No additional smearing, translation, truth selection or interpolation between generated masses is applied. Here and below, “yield” means selected-candidate yield, not a number of generated resonances.

The signal contribution in a fitted bin is Ap_i . In particular,

$$f_D = \sum_{i \in D} p_i, \quad f_W = \sum_{i \in W} p_i \quad (2)$$

remain the probabilities for a selected candidate to enter D and W . The template is normalized before either restriction, so A estimates the full selected yield N . Expected fractions and actual draws inside the support, window and training bins are saved for each toy.

m_0	$c(m_0)$	σ_m	f_D	$f_{W,P}$	$f_{W,C}$
60	58.792	2.075	0.998	0.392	0.396
80	78.103	2.288	0.999	0.675	0.709
100	97.688	2.570	0.999	0.731	0.781
120	117.319	2.920	0.999	0.745	0.803
140	136.956	3.339	0.999	0.764	0.812
160	156.658	3.827	0.999	0.751	0.816
180	176.393	4.383	0.999	0.752	0.803
200	196.182	5.008	0.998	0.756	0.792
220	215.941	5.702	0.998	0.737	0.771
240	235.617	6.465	0.997	0.722	0.748
260*	255.173	7.296	0.995	0.701	0.722

Table 2: Fixed MC centers, nominal resolutions and selected fractions. Masses, centers and resolutions are in MeV. P and C denote the pole- and MC-core-centered masks. The asterisk marks the 260 MeV extension, which uses the archived 250 MeV kernel state.

3 Extraction and uncertainty calculation

The pole-centered mask is $m_0 \pm 2.25\sigma_m(m_0)$; the shifted mask is $c(m_0) \pm 2.25\sigma_m(m_0)$. The center c is determined from MC alone using the local integrated Gaussian plus nonnegative affine pedestal fit defined in v6.1. The same definition is applied at 260 MeV. The MC shape does not move when the window moves. The resolution is

$$\sigma_m(m) = 0.00184825 - 0.001375m + 0.085875m^2, \quad m, \sigma_m \text{ in GeV.} \quad (3)$$

Kernel parameters remain anchored at m_0 , with the archived 250 MeV state used for the 260 MeV extension. The MC center and nominal width are fixed before drawing the toys.

In the primary fit the GP is trained on the injected toy's exterior bins. Its archived kernel hyperparameters remain fixed, while the count-dependent targets, noise terms, conditional mean $\hat{\mathbf{b}}$ and predictive covariance are recomputed. The v6.1 factorization retains the covariance modes represented by L , with $V = LL^T$. The inherited likelihood profiles their nuisance coordinates $\boldsymbol{\theta}$:

$$\begin{aligned} -\ln \mathcal{L}(A, \boldsymbol{\theta}) &= \sum_{i \in W} [\lambda_i - n_i \ln \lambda_i] + \frac{1}{2} \boldsymbol{\theta}^T \boldsymbol{\theta}, \\ \lambda_i &= \hat{b}_i + (L\boldsymbol{\theta})_i + Ap_i > 0. \end{aligned} \quad (4)$$

The fitted amplitude $\hat{A}$ is allowed to be negative, retaining the signed behavior around zero signal. Its uncertainty σ_A is obtained from the observed curvature of the profiled likelihood.

Two controls use the same injected counts in the fitted window. The clean-sideband control trains the GP on the background-only bins of that toy, removing the injected tail from training. The known-background control fixes the background to μ_i and has no GP nuisance parameters. The former isolates the change caused by signal in the training sample; the latter also removes background-estimation fluctuations and the GP nuisance covariance.

Writing $\ell_{\text{prof}}(A)$ for the negative log likelihood minimized over $\boldsymbol{\theta}$ at fixed A , the pull and likelihood ratio at the injected truth are

$$p = \frac{\hat{A} - N}{\sigma_A}, \quad q_N = 2 \left[\ell_{\text{prof}}(N) - \ell_{\text{prof}}(\hat{A}) \right]. \quad (5)$$

The nominal 68.27% and 95% profile sets include N when $q_N \leq 1$ and $q_N \leq \chi_1^2(0.95)$, respectively. The fraction of toys satisfying each condition is reported with an exact 95% Clopper–Pearson interval. Because the signal in Eq. (1) has an exact multinomial total while Eq. (4) uses Poisson counts, width one and the nominal inclusion fractions are reference values, including for known background.

4 Mean recovered yield

For each mass, level and method, the raw recovery ratio over $T = 40$ toys is

$$R(m_0, N) = \frac{1}{NT} \sum_{t=1}^T \hat{A}_t(m_0, N). \quad (6)$$

It includes the response to the injected signal and any fitted offset already present at zero injection. At $N = 30,000$, the ranges over 60–240 MeV are 0.822–1.011 for the pole-centered window and 0.845–1.073 for the shifted window.

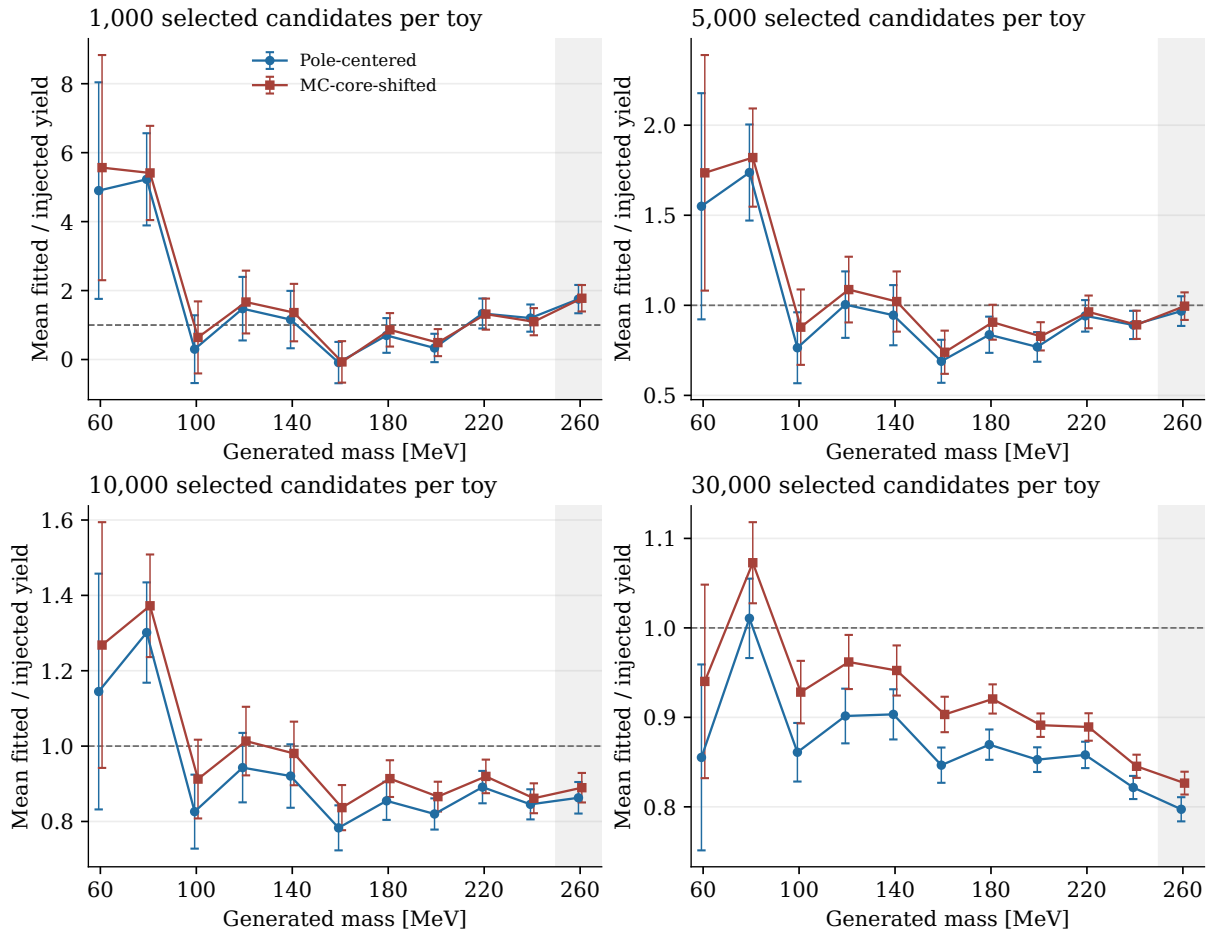


Figure 1: Mean raw recovery from the primary injected-sideband fits. Error bars are the sample standard error across 40 toys; the horizontal line is exact mean recovery. The gray band marks the separate 260 MeV extension.

At the smaller injections, the fitted background offset and its fluctuations can be comparable to N , producing large changes in the raw ratio. Section 8 subtracts each toy’s zero-injection fit to isolate the incremental response.

5 Pulls and profile inclusion

The mean pull expresses the average residual in fitted uncertainty units. A zero mean pull need not imply $E[\hat{A}] = N$, since the uncertainty σ_A can vary with the residual. The tables therefore give the mean yield and mean uncertainty alongside the pull summaries.

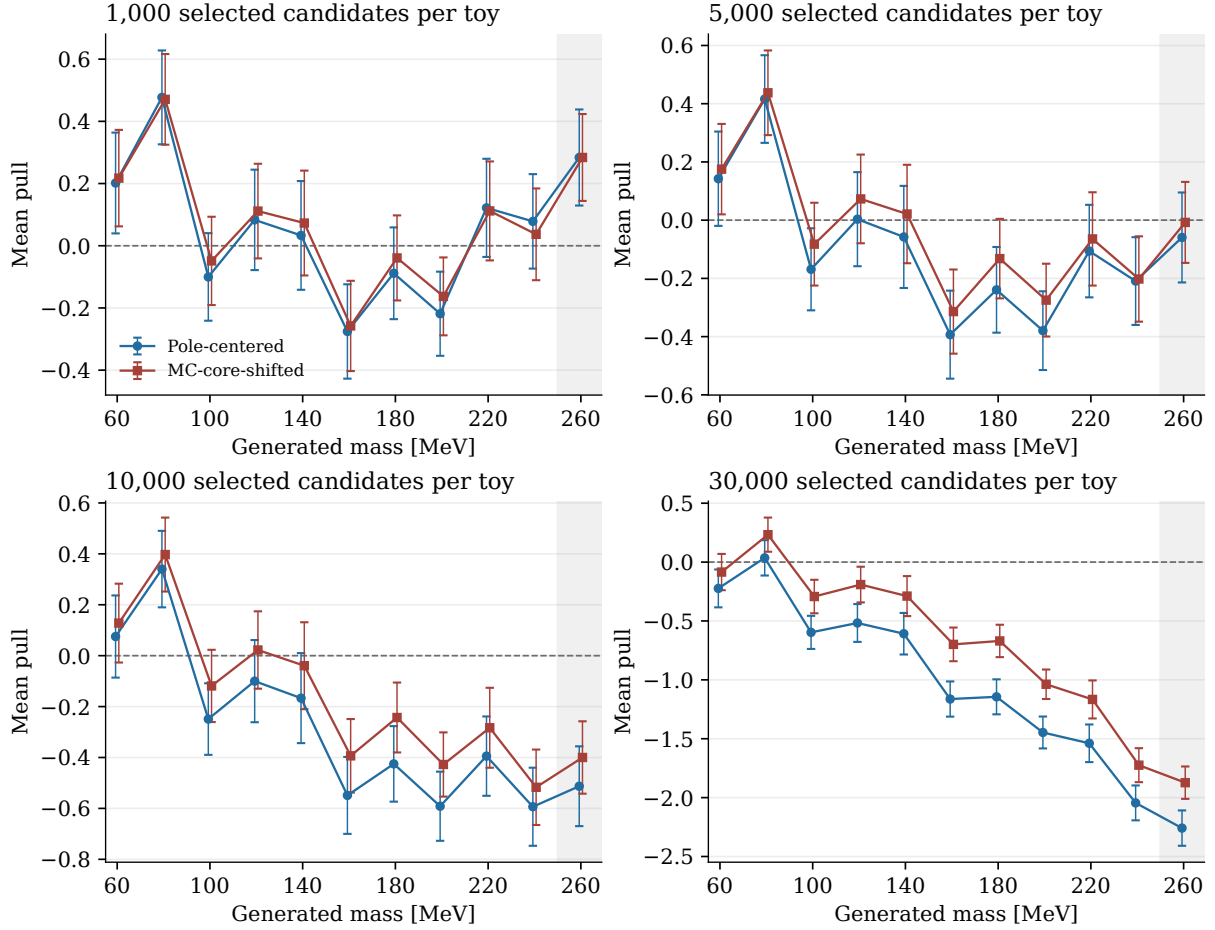


Figure 2: Mean pulls in the primary fits. Bars are the sample pull standard deviation divided by $\sqrt{40}$. Both methods share the same background and injection draws.

At 240 MeV and $N = 30,000$, the mean fitted yields are 24,649 and 25,360 candidates for the pole- and core-centered windows. The corresponding mean pulls are -2.04 and -1.72. The shift reduces the negative residual but leaves a substantial deficit.

5.1 Dispersion and interval inclusion

The pull width is the sample standard deviation, calculated with the $T - 1$ denominator. It is estimated directly from the individual pulls without fitting a Gaussian to the histograms. With 40 toys its sampling uncertainty remains appreciable: even for Gaussian pulls, the width has an approximate 11% relative standard error. The four injection levels also share background spectra, so their summaries are correlated.

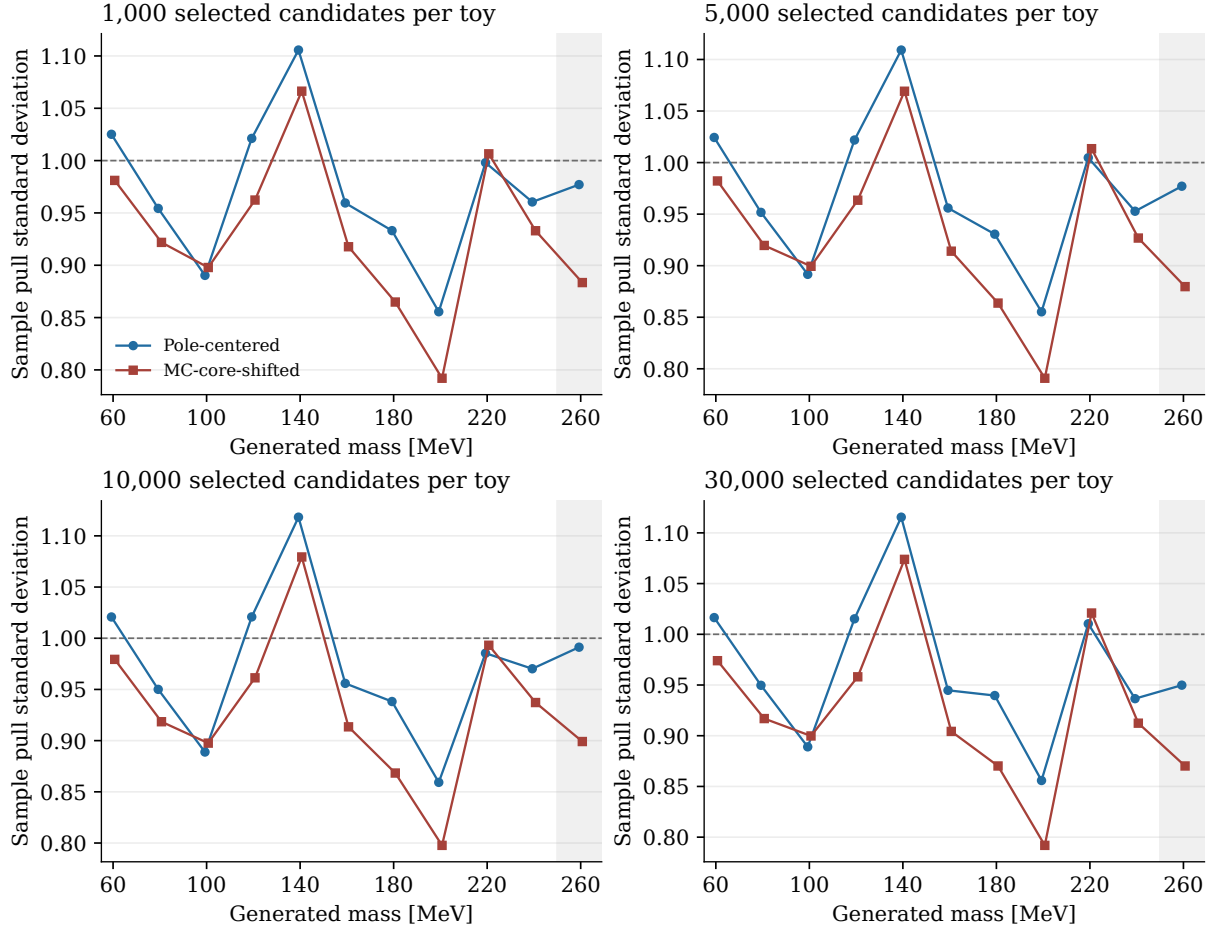


Figure 3: Sample pull widths for the primary fits. The dashed line marks one. The exact- N signal ensemble and the GP uncertainty treatment need not produce this width even in the absence of mean bias.

At 240 MeV and $N = 30,000$, the nominal 95% profile set contains the truth in 15/40 pole-centered fits and 22/40 shifted fits. The corresponding 95% binomial intervals are 0.23–0.54 and 0.38–0.71. These counts describe this toy ensemble; 40 trials do not establish calibrated coverage.

6 Zero-injection behavior

The $N = 0$ fits use the same background spectra that appear at the four nonzero injection levels. They reveal the baseline fluctuations and offsets that contribute to R at small N . Clean and injected GP sidebands coincide at zero injection, so only one GP result is needed.

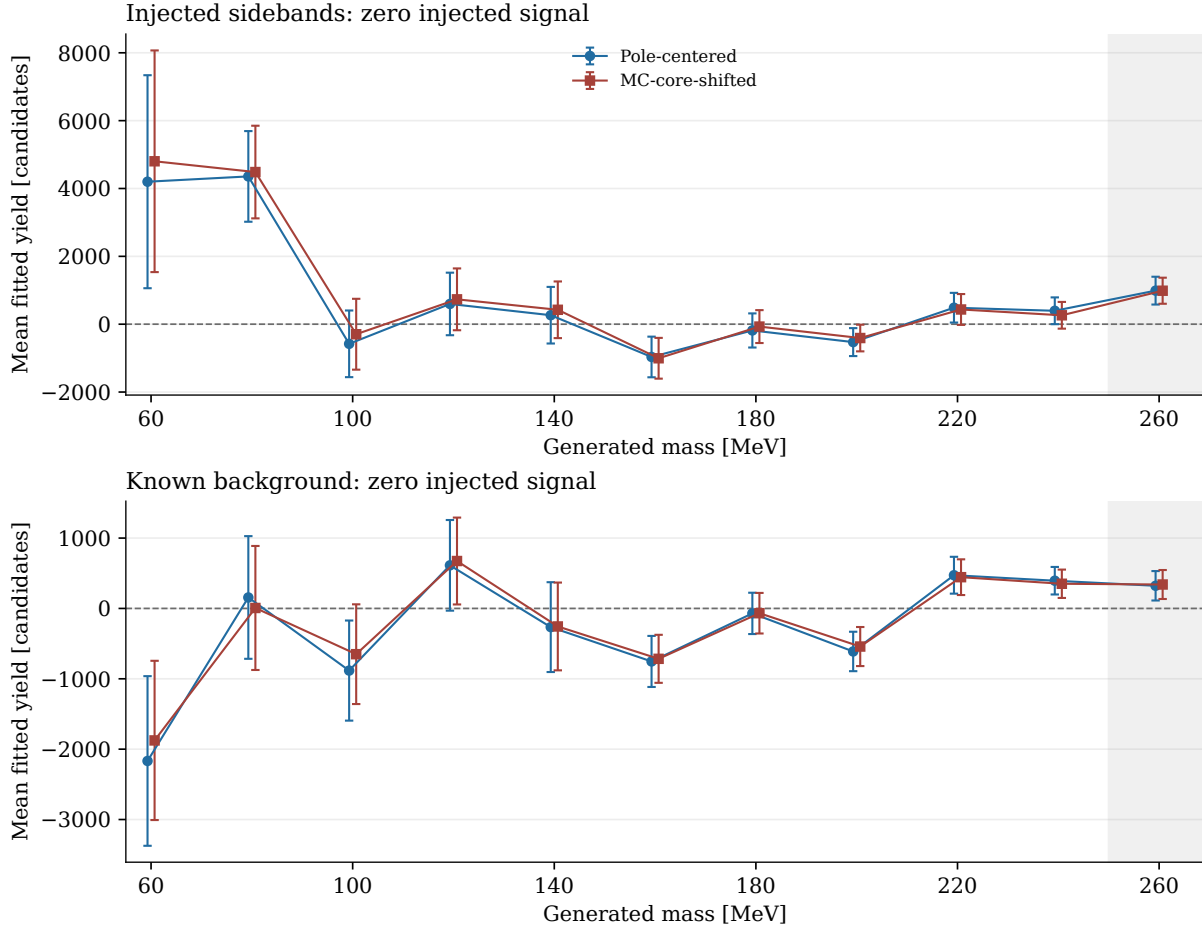


Figure 4: Mean signed fitted yield at zero injection, with standard errors across 40 background toys. The upper panel uses GP background estimation; the lower panel fixes the known generating expectation.

The uncertainty in the known-background control assumes that the background mean is known exactly. It excludes the statistical cost of estimating that mean from sideband data and cannot be substituted for the uncertainty in the GP extraction. The paired subtraction below uses each method's own zero-injection fit.

7 Signal in the training sidebands

The 30,000-candidate injections make the effect of the broad MC tails visible relative to the statistical fluctuations. Across both masks and 60–240 MeV, clean-sideband mean recovery spans 0.966–1.158, while known-background recovery spans 0.928–1.022.

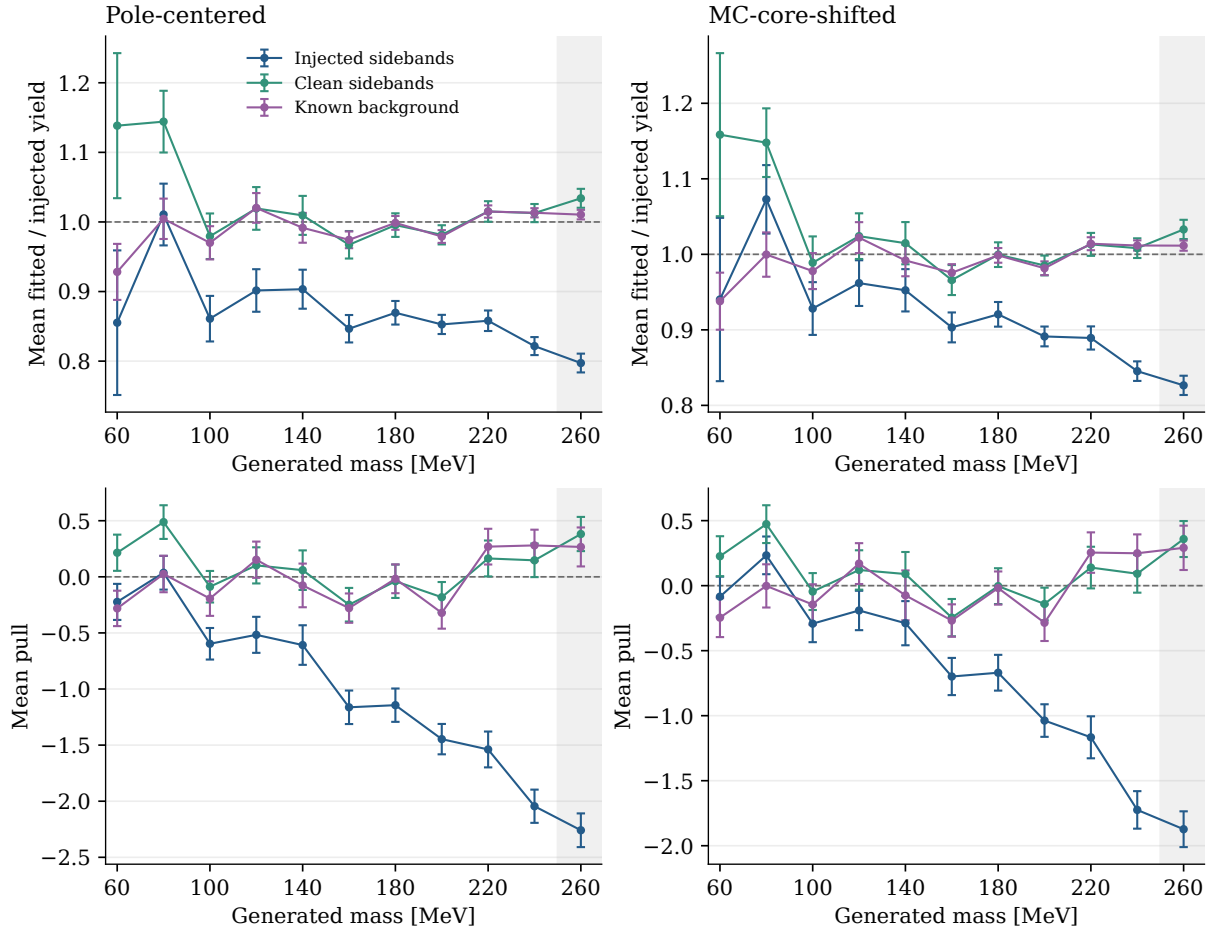


Figure 5: Recovery and mean pulls at $N = 30,000$ for the three extraction conditions. Only the training sample changes between the injected- and clean-sideband GP fits. The known-background fit additionally fixes the fitted background vector.

The effect also occurs without toy fluctuations. For a deterministic mean-count spectrum at 240 MeV with $N = 30,000$, the extracted yields are 24,023/24,859 with injected sidebands, 29,743/29,724 with clean sidebands and 30,000/30,000 with known background, where each pair lists pole-centered/core-centered results. The additional sideband counts raise the background prediction in the signal window, leaving less yield for the signal component. Centering the mask on the MC core reduces this loss but does not remove the broad tail from the training sample.

8 Paired response to the injected signal

To separate the response to additional signal from the zero-injection offset, define

$$R_{\text{inc}}(m_0, N) = \frac{1}{NT} \sum_{t=1}^T \left[\hat{A}_t(m_0, N) - \hat{A}_t(m_0, 0) \right]. \quad (7)$$

This quantity measures the change in fitted yield caused by adding signal to the same background toy. The raw estimator and its pulls retain the zero-injection offset.

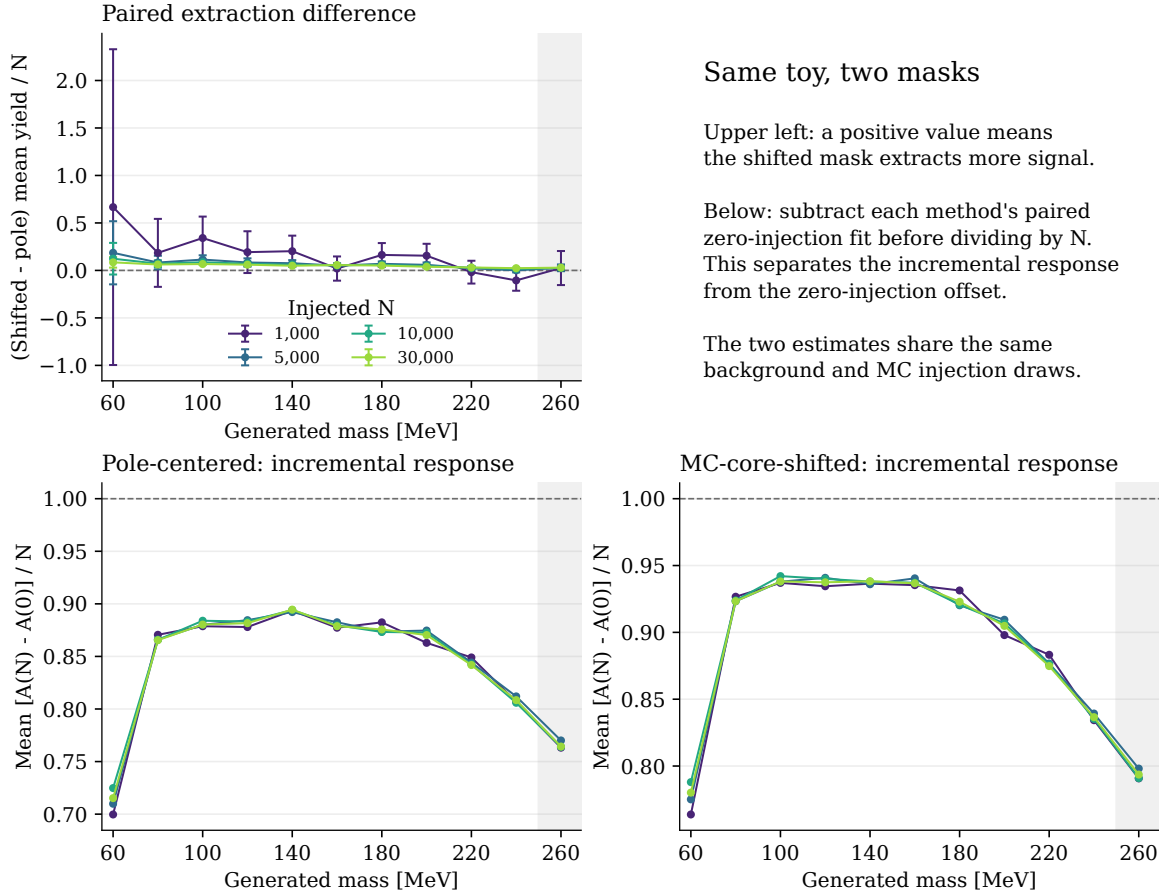


Figure 6: The upper panel compares the two fitted yields within each identical toy, divided by N . Error bars use the standard error of that paired difference. Lower panels show R_{inc} for each window prescription.

At $N = 30,000$, R_{inc} spans 0.715–0.895 for the pole-centered mask and 0.780–0.938 for the shifted mask over 60–240 MeV. Adding the second set of 20 toys changes this increment by at most 0.29 percentage points across those masses and methods. The first 20 draws and their fit results are unchanged and are preserved in the archived 20-toy release. The ensemble contains 1,760 injected toys and 440 background toys, giving 3,520 primary nonzero-injection fits and 12,320 fits including the controls. The persistent deficit in the paired response agrees with the clean-sideband and mean-count comparisons.

A Per-mass pull distributions

Each outline below contains 40 primary-fit pulls. The methods share bin edges within a mass panel, while ranges may differ between masses. Histograms show counts rather than a smooth density estimate.

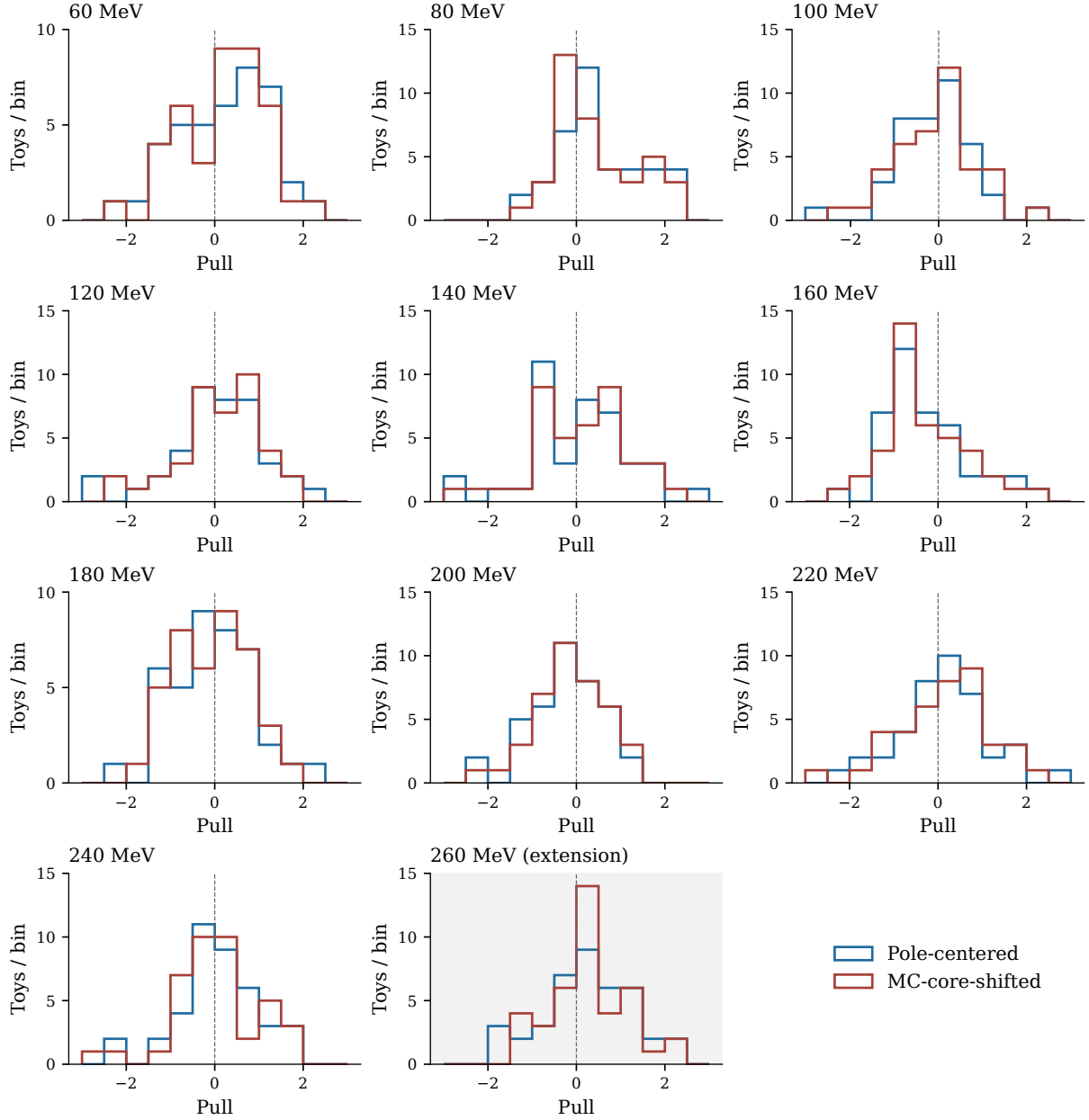


Figure 7: Pulls for $N = 1,000$. The shaded 260 MeV panel denotes the extraction extension.

A.1 Five thousand injected candidates

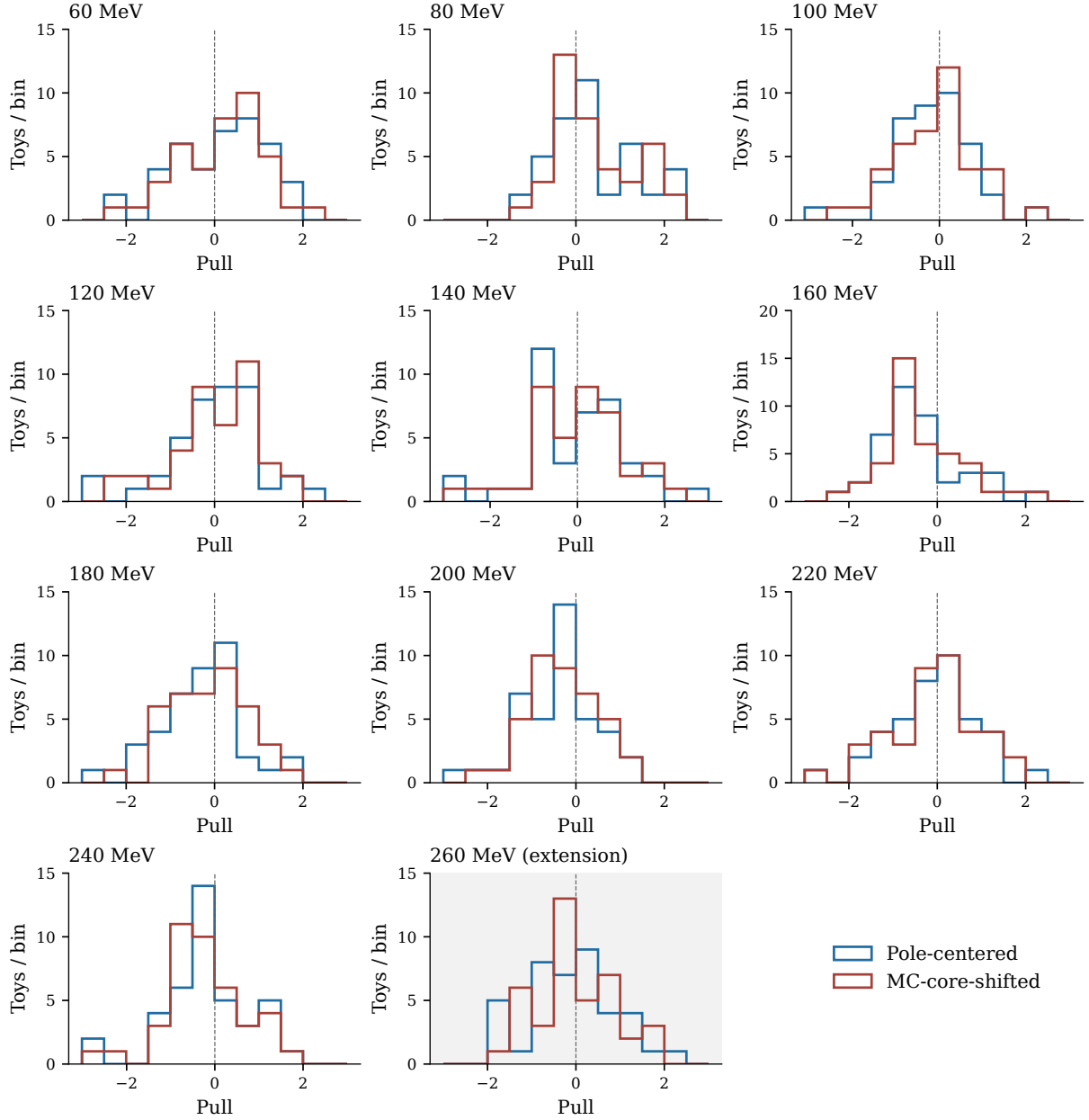


Figure 8: Pulls for $N = 5,000$; 40 toys per method and mass.

A.2 Ten thousand injected candidates

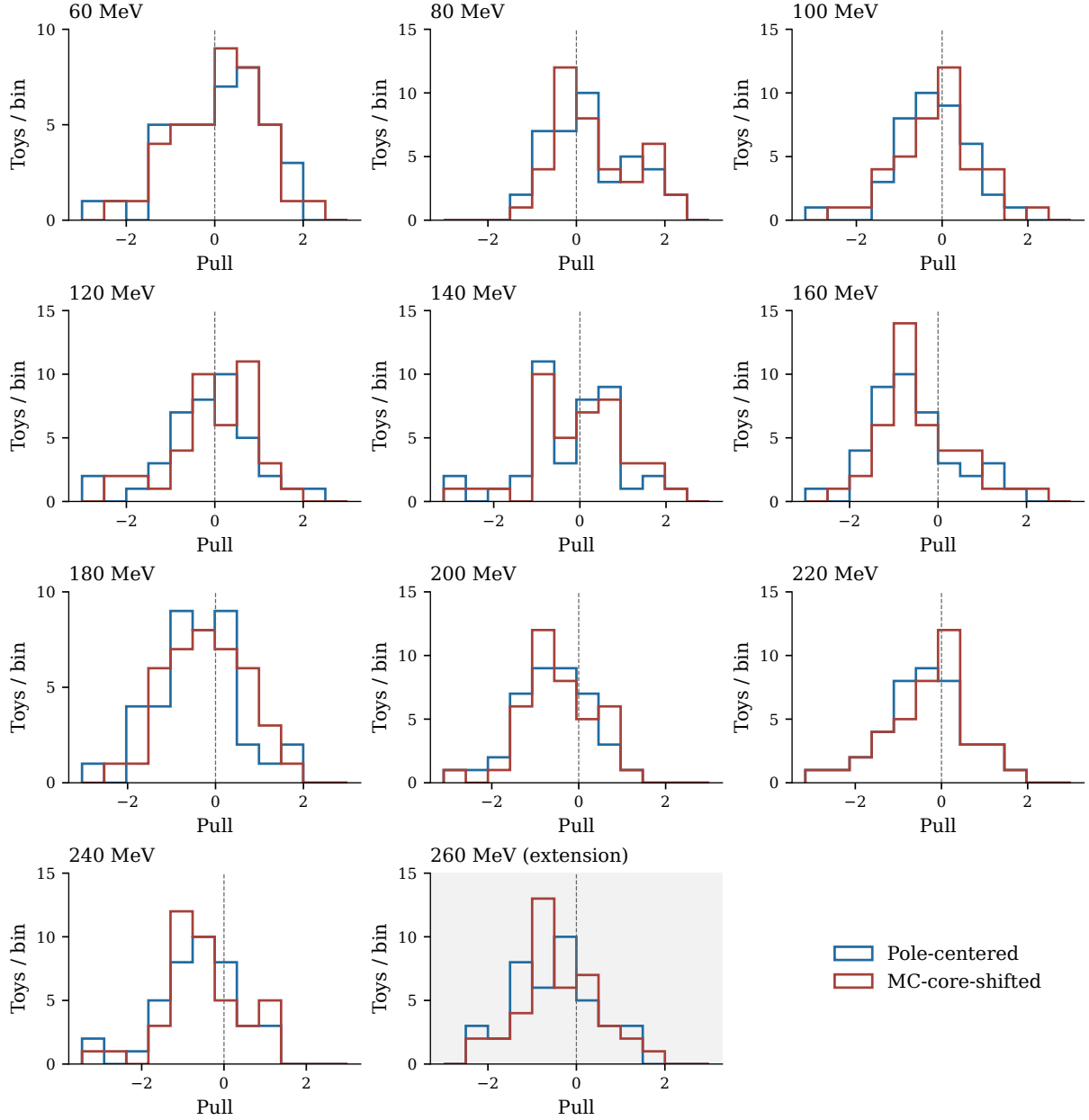


Figure 9: Pulls for $N = 10,000$; 40 toys per method and mass.

A.3 Thirty thousand injected candidates

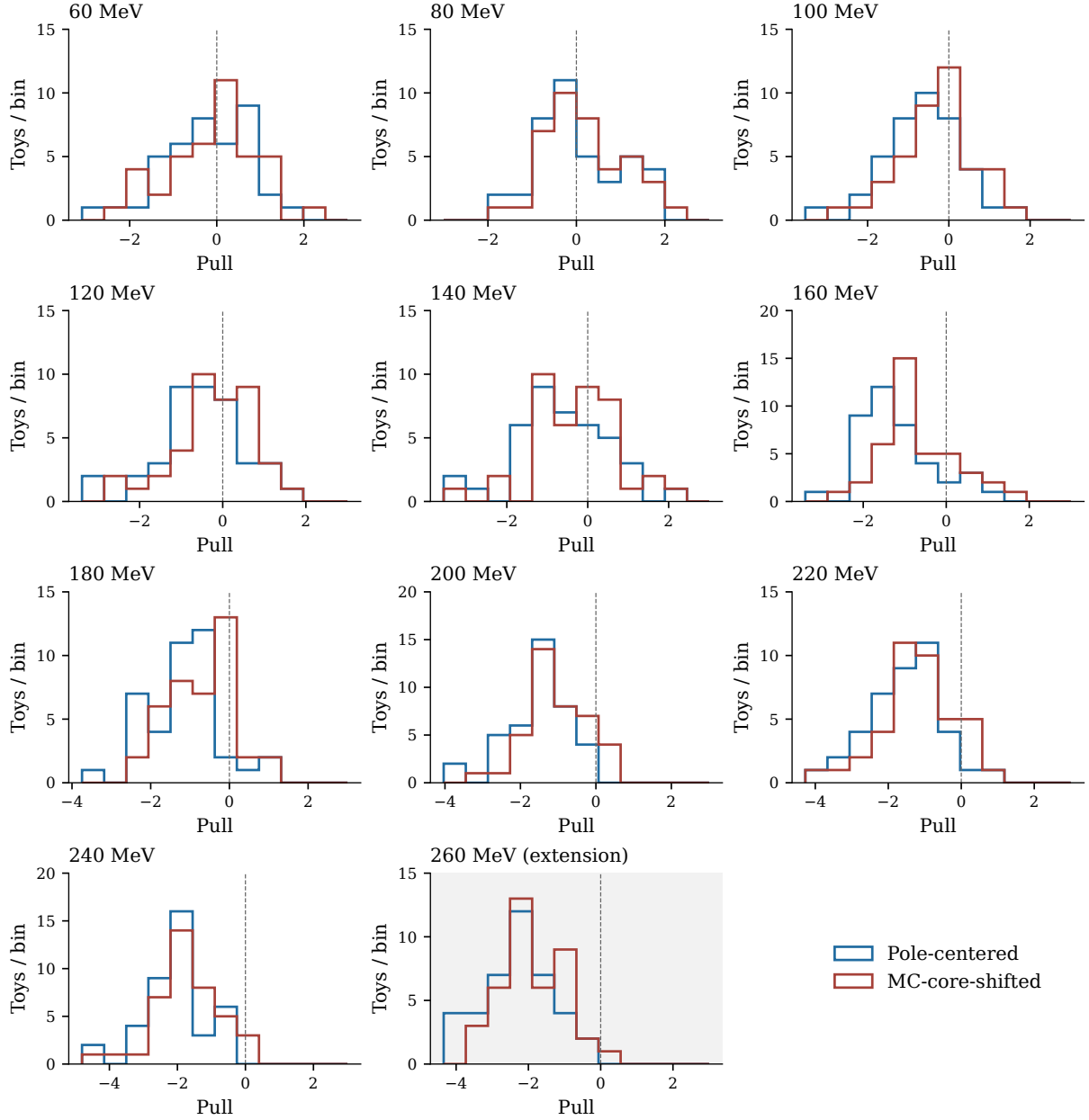


Figure 10: Pulls for $N = 30,000$; 40 toys per method and mass.

B Primary-fit numerical results

Tables below contain every primary mass/injection/method cell. P denotes the pole-centered window and C the MC-core-centered window. The mass m_0 is in MeV; $\widehat{A}$ and $\overline{\sigma_A}$ are in selected candidates. The raw recovery is R , and $\bar{p}$ and s_p are the mean and sample standard deviation of the pulls. The last two columns give the nominal 68.27% and 95% profile-inclusion counts out of 40.

m_0	Mask	$\widehat{A}$	$\overline{\sigma_A}$	R	$\bar{p}$	s_p	k_{68}	k_{95}
60	P	4,900	19,376	4.900	0.20	1.03	24	38
60	C	5,566	21,052	5.566	0.22	0.98	27	38
80	P	5,227	8,861	5.227	0.48	0.95	26	36
80	C	5,411	9,374	5.411	0.47	0.92	28	36
100	P	300	6,983	0.300	-0.10	0.89	33	38
100	C	641	7,360	0.641	-0.05	0.90	29	38
120	P	1,475	5,708	1.475	0.08	1.02	29	37
120	C	1,667	5,985	1.667	0.11	0.96	29	38
140	P	1,158	4,761	1.158	0.03	1.11	29	37
140	C	1,361	4,942	1.361	0.07	1.07	29	37
160	P	-89	3,950	-0.089	-0.28	0.96	27	38
160	C	-69	4,147	-0.069	-0.26	0.92	29	38
180	P	698	3,412	0.698	-0.09	0.93	29	38
180	C	861	3,549	0.861	-0.04	0.86	30	40
200	P	335	3,041	0.335	-0.22	0.86	31	38
200	C	490	3,132	0.490	-0.16	0.79	32	39
220	P	1,335	2,752	1.335	0.12	1.00	29	38
220	C	1,317	2,833	1.317	0.11	1.01	27	38
240	P	1,202	2,596	1.202	0.08	0.96	30	38
240	C	1,097	2,669	1.097	0.04	0.93	29	38
260*	P	1,752	2,659	1.752	0.28	0.98	25	38
260*	C	1,777	2,745	1.777	0.28	0.88	27	37

Table 3: Primary fits for $N = 1,000$. The asterisk marks 260 MeV.

The 260 MeV extension uses the archived 250 MeV kernel and is excluded from the 60–240 MeV summary ranges. Full-precision results, all controls, Monte Carlo standard errors and Clopper–Pearson intervals are retained in `results/summary.csv`; individual fits are in `results/toys.csv`. Input identities and the fixed construction are recorded in `protocol.json` and the provenance files.

B.1 Five thousand injected candidates

m_0	Mask	$\widehat{A}$	$\overline{\sigma_A}$	R	$\bar{p}$	s_p	k_{68}	k_{95}
60	P	7,749	19,377	1.550	0.14	1.02	25	37
60	C	8,678	21,054	1.736	0.18	0.98	28	38
80	P	8,686	8,861	1.737	0.42	0.95	26	36
80	C	9,101	9,375	1.820	0.44	0.92	28	36
100	P	3,822	6,984	0.764	-0.17	0.89	33	38
100	C	4,394	7,361	0.879	-0.08	0.90	29	38
120	P	5,019	5,709	1.004	0.00	1.02	31	37
120	C	5,437	5,986	1.087	0.07	0.96	30	38
140	P	4,726	4,762	0.945	-0.06	1.11	30	37
140	C	5,105	4,943	1.021	0.02	1.07	30	37
160	P	3,446	3,951	0.689	-0.39	0.96	26	38
160	C	3,698	4,148	0.740	-0.31	0.91	30	38
180	P	4,183	3,413	0.837	-0.24	0.93	29	39
180	C	4,531	3,550	0.906	-0.13	0.86	29	39
200	P	3,845	3,043	0.769	-0.38	0.86	28	38
200	C	4,139	3,134	0.828	-0.27	0.79	31	39
220	P	4,707	2,755	0.941	-0.11	1.00	28	37
220	C	4,817	2,835	0.963	-0.06	1.01	26	39
240	P	4,455	2,598	0.891	-0.21	0.95	29	38
240	C	4,459	2,672	0.892	-0.20	0.93	30	38
260*	P	4,839	2,664	0.968	-0.06	0.98	28	38
260*	C	4,977	2,750	0.995	-0.01	0.88	28	40

Table 4: Primary fits for $N = 5,000$; 40 toys per row. Column definitions are given at the start of Appendix B.

B.2 Ten thousand injected candidates

m_0	Mask	$\widehat{A}$	$\overline{\sigma_A}$	R	$\bar{p}$	s_p	k_{68}	k_{95}
60	P	11,448	19,379	1.145	0.08	1.02	25	38
60	C	12,682	21,056	1.268	0.13	0.98	27	38
80	P	13,014	8,862	1.301	0.34	0.95	27	36
80	C	13,724	9,376	1.372	0.40	0.92	28	38
100	P	8,261	6,985	0.826	-0.25	0.89	30	39
100	C	9,125	7,362	0.913	-0.12	0.90	30	38
120	P	9,429	5,710	0.943	-0.10	1.02	30	37
120	C	10,133	5,987	1.013	0.02	0.96	31	38
140	P	9,206	4,763	0.921	-0.17	1.12	29	36
140	C	9,805	4,944	0.981	-0.04	1.08	30	37
160	P	7,830	3,952	0.783	-0.55	0.96	22	39
160	C	8,367	4,149	0.837	-0.39	0.91	28	38
180	P	8,548	3,415	0.855	-0.43	0.94	27	39
180	C	9,137	3,552	0.914	-0.24	0.87	28	39
200	P	8,199	3,045	0.820	-0.59	0.86	28	38
200	C	8,659	3,136	0.866	-0.43	0.80	31	39
220	P	8,912	2,757	0.891	-0.39	0.99	27	37
220	C	9,196	2,838	0.920	-0.28	0.99	28	38
240	P	8,454	2,602	0.845	-0.59	0.97	28	38
240	C	8,616	2,676	0.862	-0.52	0.94	27	38
260*	P	8,629	2,669	0.863	-0.51	0.99	24	36
260*	C	8,896	2,755	0.890	-0.40	0.90	29	38

Table 5: Primary fits for $N = 10,000$; 40 toys per row. Column definitions are given at the start of Appendix B.

B.3 Thirty thousand injected candidates

m_0	Mask	$\widehat{A}$	$\overline{\sigma_A}$	R	$\bar{p}$	s_p	k_{68}	k_{95}
60	P	25,657	19,387	0.855	-0.22	1.02	28	38
60	C	28,206	21,064	0.940	-0.08	0.97	28	38
80	P	30,320	8,866	1.011	0.04	0.95	27	40
80	C	32,185	9,379	1.073	0.23	0.92	29	39
100	P	25,830	6,989	0.861	-0.60	0.89	27	39
100	C	27,846	7,365	0.928	-0.29	0.90	30	39
120	P	27,046	5,713	0.902	-0.52	1.02	27	37
120	C	28,857	5,991	0.962	-0.19	0.96	29	38
140	P	27,100	4,768	0.903	-0.61	1.12	21	36
140	C	28,571	4,949	0.952	-0.29	1.07	26	36
160	P	25,397	3,958	0.847	-1.16	0.94	13	33
160	C	27,097	4,155	0.903	-0.70	0.90	21	38
180	P	26,085	3,422	0.870	-1.14	0.94	19	32
180	C	27,617	3,559	0.921	-0.67	0.87	25	38
200	P	25,582	3,054	0.853	-1.45	0.86	11	29
200	C	26,738	3,144	0.891	-1.04	0.79	18	37
220	P	25,741	2,768	0.858	-1.54	1.01	9	28
220	C	26,677	2,849	0.889	-1.17	1.02	20	32
240	P	24,649	2,616	0.822	-2.04	0.94	6	15
240	C	25,360	2,690	0.845	-1.72	0.91	8	22
260*	P	23,918	2,692	0.797	-2.26	0.95	4	15
260*	C	24,795	2,778	0.826	-1.87	0.87	5	21

Table 6: Primary fits for $N = 30,000$; 40 toys per row. Column definitions are given at the start of Appendix B.